

```
apiVersion: v1
kind: LimitRange
metadata:
  name: resource-limits
  namespace: dev
spec:
  limits:
  - type: Container
    default:
      cpu: "500m"
      memory: "512Mi"
    defaultRequest:
      cpu: "250m"
      memory: "256Mi"
```

```
apiVersion: v1
kind: LimitRange
metadata:
  name: enforce-boundaries
  namespace: production
spec:
  limits:
  - type: Container
    min:
      cpu: "200m"
      memory: "256Mi"
    max:
      cpu: "1"
      memory: "1Gi"
```

```
apiVersion: v1
kind: LimitRange
metadata:
  name: pvc-storage-limits
  namespace: dev-namespaces
spec:
  limits:
  - type: PersistentVolumeClaim
    min:
      storage: 1Gi
    max:
      storage: 10Gi
```

```
apiVersion: v1
kind: ResourceQuota
metadata:
  name: compute-quota
  namespace: dev-namespaces
spec:
  hard:
    requests.cpu: "2"
    requests.memory: "4Gi"
    limits.cpu: "4"
    limits.memory: "8Gi"
    pods: "10"
    persistentvolumeclaims: "5"
    requests.storage: "50Gi"
```

```
kubectl apply -f compute-quota.yaml
```

```
kubectl get resourcequota -n <namespace>
kubectl describe resourcequota <quota-name> -n <namespace>
```

- kube_resourcequota
- kube_resourcequota_usage
- kube_resourcequota_hard

```
expr: kube_resourcequota_usage / kube_resourcequota_hard > 0.9
```

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
podPidsLimit: 100
```

```
kubelet --pod-max-pids=100
```

```
cat /sys/fs/cgroup/pids/kubepods.slice/kubepods-<pod_id>/pids.max
```

```
docker inspect <container_id> | grep PidsLimit
```

```
systemReserved:
  cpu: "500m"
  memory: "512Mi"
```

```
kubeReserved:
  cpu: "200m"
  memory: "300Mi"
```

```
evictionHard:
  memory.available: "100Mi"
  nodefs.available: "10%"
```

```
apiVersion: kubelet.config.k8s.io/v1beta1
kind: KubeletConfiguration
systemReserved:
  cpu: "500m"
  memory: "512Mi"
kubeReserved:
  cpu: "250m"
  memory: "256Mi"
evictionHard:
  memory.available: "100Mi"
```

```
affinity:
  nodeAffinity:
    requiredDuringSchedulingIgnoredDuringExecution:
      nodeSelectorTerms:
        - matchExpressions:
            - key: "disktype"
              operator: "In"
              values:
                - ssd
```

```
kubectl taint nodes node1 key=value:NoSchedule
```

```
tolerations:
  - key: "key"
    operator: "Equal"
    value: "value"
    effect: "NoSchedule"
```

```
resources:
  requests:
    memory: "256Mi"
    cpu: "250m"
  limits:
    memory: "512Mi"
    cpu: "500m"
```